



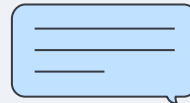
Natural Language processing

Group 17

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Polite Guard Dataset

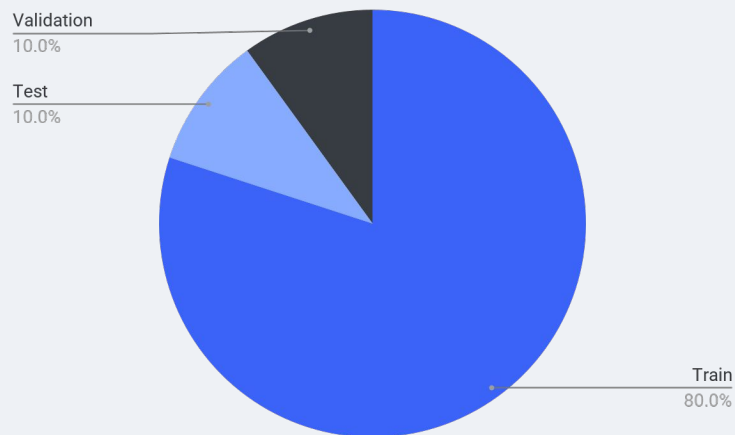
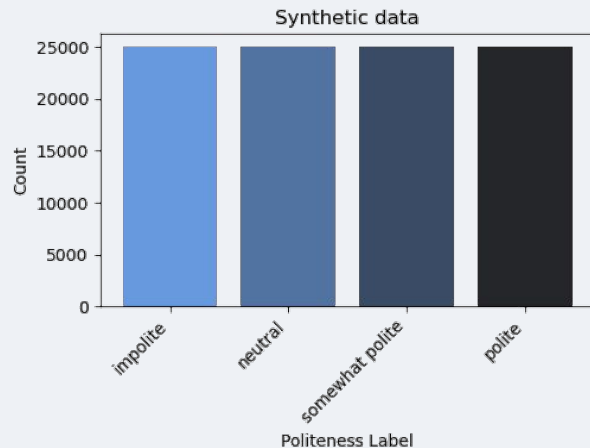
Goal: Classification of text into **polite**, **somewhat polite**, **neutral** or **impolite**

Synthetic dataset: 100k samples generated by LLMs, 200 real annotated examples with personal identifiers removed

Key Considerations: No personal identifiers, even distribution of classes (25% of each class in every split)

Performance Benchmark (as reported on the [official dataset webpage](#)):

- Accuracy: 92%
- F1-Score: 92%



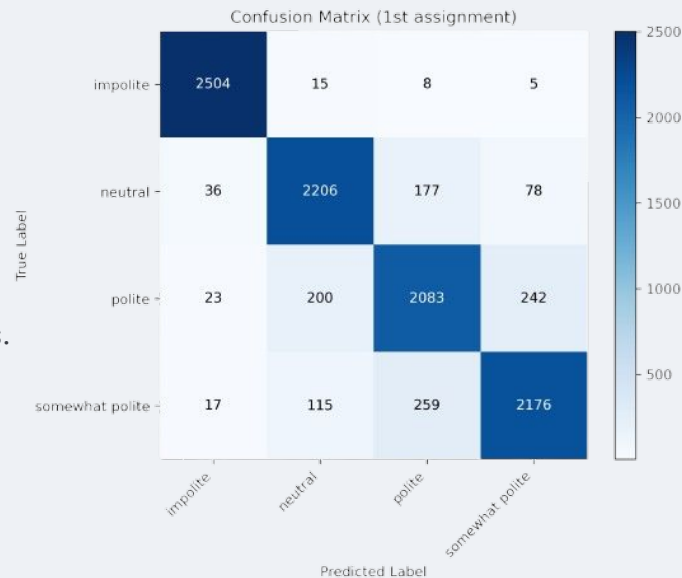
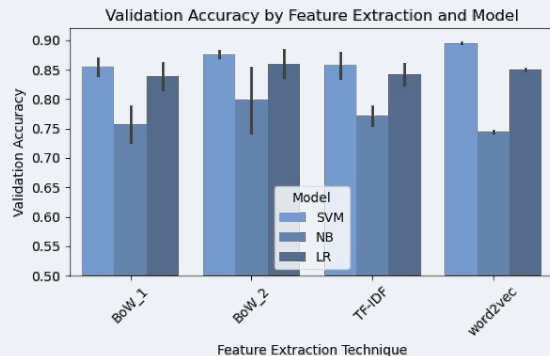
1st assignment - experiments and best results

Models: Naive Bayes, Logistic Regression and SVM

Hyperparameter Tuning: Different parameters and feature extraction techniques tested for each model

Overall Performance

- SVM + Word2Vec achieved **88.48% accuracy** and **88.5% F1-Score** and balanced performance across politeness levels.
- Intra-class metrics show better performance for “lower” politeness labels (class impolite)



Model Choices and Fine Tuning

Baseline Models

(1) BERT (bert-base-uncased)

- **Architecture Fit:** Encoder-only architecture is ideal to understand nuanced phrases and context
- **Relevant Pretraining:** pretrained on large, diverse English corpora, closely matching the domain and tone of our dataset
- **Proven Performance:** widely used across NLP, showing good results in sequence classification. Base for the state-of-the-art model

(2) RoBERTa (RoBERTa-base)

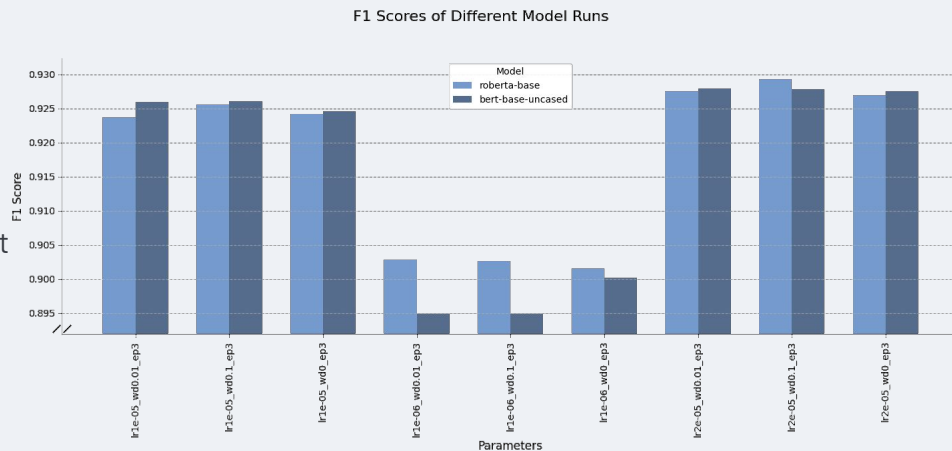
- An improvement over BERT
- Not uncased, so may capture more information

Hyperparameter Tuning

- Tuned hyperparameters (learning rate, weight decay)
- Selected best configs using F1-score on validation set and trained for 10 epochs:

Bert: lr 2e-5; wd 0.01 | **RoBERTa:** lr 2e-5; wd 0.1

Parameter	Values
Model	bert-base-uncased, roberta
Epochs	10
Learning rate	1e-6, 1e-5, 2e-5
Weight decay	0.0, 0.01, 0.1



Domain Adaptation and Parameter Efficient Fine Tuning (LoRA and Ia3)

Domain Adaptation

- Added intermediate fine-tuning via Masked Language Modeling (MLM)
- Adapted models to domain-specific language using target corpus

MLM Classification & Evaluation

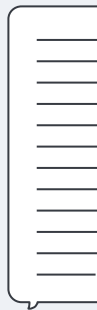
- Applied adapted models to classification tasks
- Tracked training with perplexity
- Compared performance gains over non-adapted baselines

LoRA (Low-Rank Adaptation)

- Freezes pre-trained weights, adds trainable low-rank matrices

Ia3 (Infused Adapter by Inhibiting and Amplifying Inner Activations)

- Rescales model activations with learned vectors



Prompting Language Models

We compared the performance of various **models** and **prompt styles**, including zero-shot, few-shot, and chain of thought approaches.

Models tested: Llama 3-8B, Mistral-7B-instruct-v0.3, Gemini 2.0 Flash

Our findings show that chain of thought reasoning has the potential to significantly enhance results.

Prompt Type	Description	Example
Zero shot	Class names	Instruction: Analyze the following text and classify its politeness level... Respond with only one of the category names... Input Text: {text} Classification:
Few shot	Class names + examples	Instruction: The classification must be one of the following... Examples... Input Text: {text} Classification:
Few shot + chain of shot	Class names + examples + reasoning	Instruction: Classify text as polite, somewhat polite, neutral, or impolite. First, briefly explain your reasoning for the classification based on the definitions provided below. Examples: "Happy to accommodate..." -> Polite Reasoning: Classification:

Experimental results

Best Model & Hyperparameters With Domain Adaptation

Model: RoBERTa (Same classifier hyperparameter + MLM)

Learning Rate: $2e-5$ Mask Probability: 0.1

Weight Decay: 0.1 Epochs: 2

Test Accuracy: **91.90%** Perplexity (on MLM Data): 1.49

Test F1-Score: 91.87% Test Accuracy: **92.40%**

Test F1-Score: 92.37%

- Both LoRA and IA3 significantly reduce training time
- LoRA achieves a much smaller drop in performance, presenting a better tradeoff

Model	Method	Δ F1 (vs Base)	Δ Time (vs Base)
BERT	LoRA	-2.25%	-26.16%
	IA3	-15.82%	-26.03%
RoBERTa	LoRA	-1.91%	-25.33%
	IA3	-7.35%	-24.42%

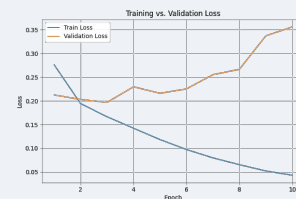
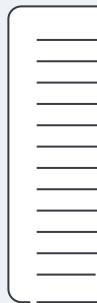


Fig. Loss curves for Bert(left) and RoBERTa(Right)



Experimental results

LLM Prompting Results (800 samples subset)

- LLM performance (~60-70% F1) significantly lower than fine-tuned models (92%+ F1)

Prompt Strategy Results:

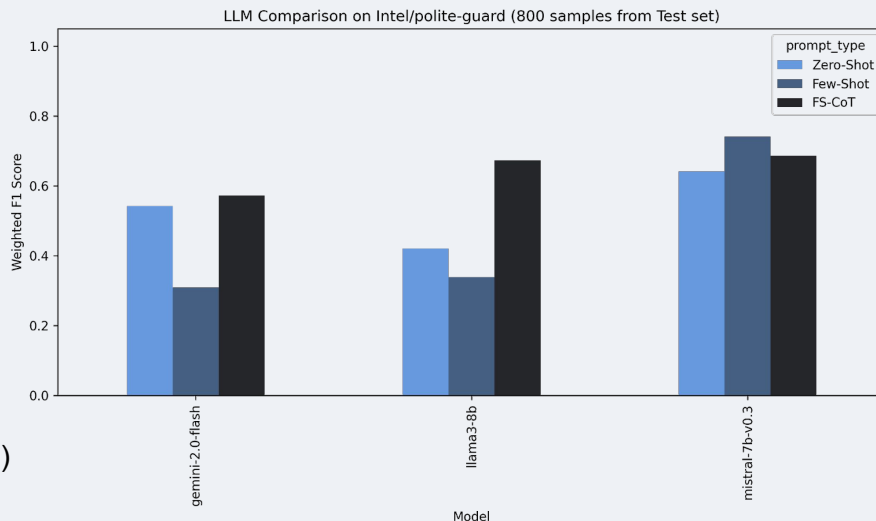
- Few-shot Chain-of-Thought generally outperformed zero-shot and few-shot approaches
- CoT reasoning helps but doesn't close the performance gap

Surprising Model Performance:

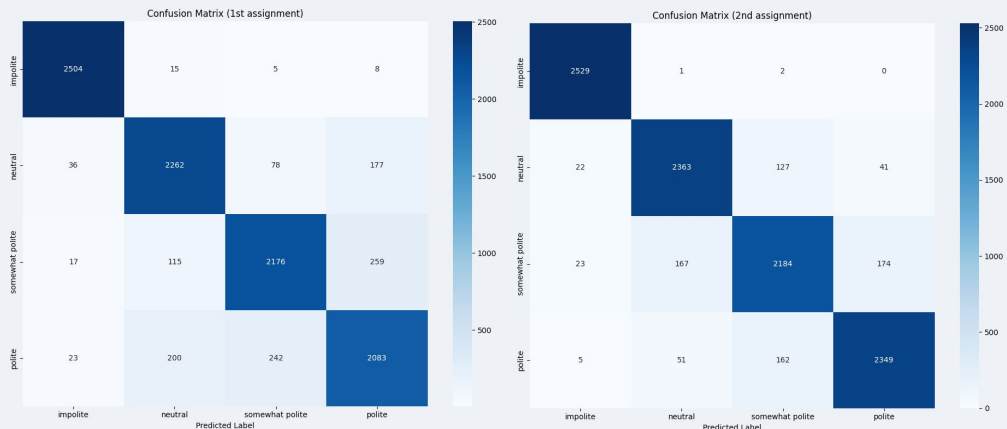
- Gemini 2.0 Flash performed unexpectedly poorly compared to smaller models (Llama 3-8B, Mistral-7B)

Hypothesis:

- LLM-generated dataset creates ambiguous class boundaries that even humans would struggle with. Fine-tuned models better capture these synthetic data patterns



Error Analysis

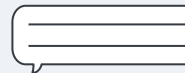


Class	Correct predictions	F1-Score
Impolite	+25	+0.99%
Neutral	+101	+4.10%
Somewhat Polite	+8	+1.08%
Polite	+266	+9.47%

What is the class of the following text?

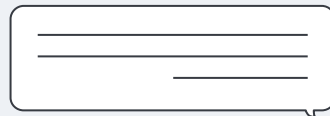
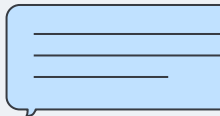
"It's a nice awareness that i learnt about this"

True label	Predicted
Polite	Somewhat Polite



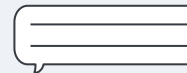
Conclusions

- RoBERTa + Domain Adaptation: 92.40% F1-Score (matches state-of-the-art)
- LoRA: Best efficiency tradeoff (-1.91% F1, -25% training time)
- Fine-tuned models significantly outperform LLMs (92% vs 60-70% F1)
- Traditional ML remains competitive (SVM: 88.5% F1 vs RoBERTa with D.A.: 92%+)
- The synthetic nature of this dataset created fuzzy boundaries that questions the real-world applicability of these results, especially for sensitive applications like content moderation where clear, reliable classifications are crucial.



References

- Polite Guard Dataset
- Polite Guard state-of-the-art model
- Other NLP projects and sources:
 - https://github.com/kk7nc/Text_Classification?tab=readme-ov-file
 - https://github.com/lpq29743/text_classification
 - <https://github.com/Christophe-pere/Text-classification/tree/master>
 - https://edumunozsala.github.io/BlogEms/jupyter/nlp/classification/embeddings/python/2020/08/15/Intro_NLP_WordEmbeddings_Classification.html



Extra: Cross-validation results

Advantages:

- Improve robustness of performance estimations
- Assess model stability
- Better use of data

Disadvantages:

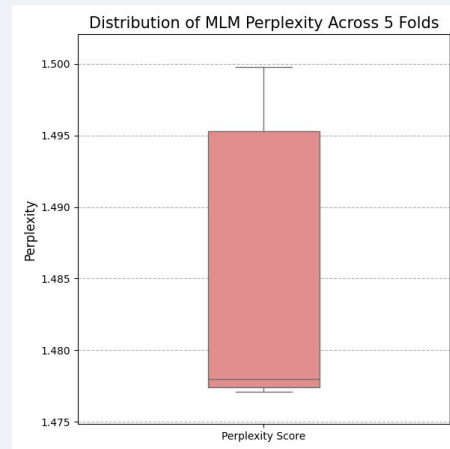
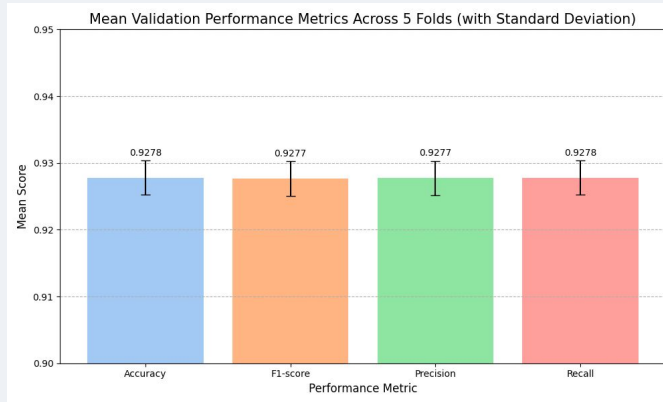
- Considerable increase in computational cost

Our solution:

- Apply cross-validation only to our best model: **RoBERTa with domain adaptation**, to confirm consistent performance

Key considerations:

- Stratified splitting
- Prevent data leakage: cannot pretrain with MLM only once
 - Perform MLM independently for each fold



Extra: Zero-Shot Prompt

<instruction>

Analyze the following text and classify its politeness level.

The politeness categories are:

polite, somewhat polite, neutral, impolite

Respond with only one of the category names listed above and place your answer under the <classification> tag.

<\instruction>

Input: {text}

<classification>



Extra: Few-Shot Prompt

<instruction>

Analyze the following text and classify its politeness level.

The politeness categories are:

polite, somewhat polite, neutral, impolite

Respond with only one of the category names listed above and place your answer under the <classification> tag.

Example 1

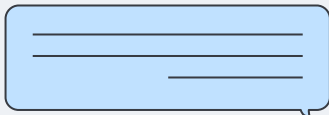
Input: "Your flight has been rescheduled for 10:00 AM tomorrow. Please check the airport's website for any updates or changes."

<classification> neutral </classification>

<\instruction>

Input: {text}

<classification>



Extra: Few-Shot Chain-of-Thought Prompt

<instruction>

Analyze the following text and classify its politeness level. First, briefly explain your reasoning for the classification based on the definitions provided below. Then, state the final label".

The politeness categories are:

polite, somewhat polite, neutral, impolite

Respond with only one of the category names listed, include both the reasoning and the final label inside the '<reasoning>' and '<classification>' tags respectively.

Example 1

Input: "Your flight has been rescheduled for 10:00 AM tomorrow. Please check the airport's website for any updates or changes."

<classification> neutral </classification>

<\instruction>

Input: {text}

<reasoning></reasoning>

<classification></classification>

